Graduation project 2025/26

DIY Hexapod robot

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Abstract

This document presents the design and construction of a remotely controlled hexapod robot. The paper identifies and analyzes challenges inherent to the development of such a system. It is structured for both direct replication of the proposed design or adaptation for alternative implementations.

1 Introduction

Legged robots, particularly hexapods, are important in robotics due to their stability and ability to traverse diverse terrains using six legs. Hexapods have application in hazardous environments and search operations, where their multi-legged configuration provides superior stability and mobility. Although this project is undertaken as a hobby, the robot serves as a representative platform for learning the fundamental principles of robotic design. This work was undertaken as a graduation project at *Smíchovská průmyslová škola a gymnázium* in the 2025/26 academic year

This project was selected to explore and learn the fundamentals of robotic design and provide practical experience in robotics, integrating multiple skills including CAD modeling, circuit design and assembly, and programming of a robotic system

The development of a hexapod robot presents several engineering challenges. From a mechanical perspective, the multi-legged configuration introduces significant structural and geometric complexity, as each leg presents multiple degrees of freedom that must remain precisely aligned to ensure stable motion. The coordination of numerous, eighteen in case of a hexapod, actuators further increases system complexity, requiring accurate timing, reliable signal distribution, and sufficient power delivery under dynamic load conditions. Finally, the objective of creating a system that is reproducible and accessible within a DIY context imposes additional constraints on component selection, cost, and overall design methodology.

The scope of this project is limited to the design, construction, and implementation of a remotely controlled hexapod robot that is intended for stable movement on flat and moderately uneven surfaces. The work focuses on mechanical design, actuator integration, power distribution, and gait control that is able to achieve reliable manual operation. Autonomous navigation, advanced terrain adaptation, computer vision, and other more advanced systems are outside the scope of this project. The system is developed within the constraints of accessible components, limited budget, and tools commonly available in an educational environment.

The remainder of this document is organized according to the development process. The initial chapter describes the modeling of the robot in a 3D modeling software. This is followed by the 3D printing process and the mechanical assembly of the structure. Following chapters address the realization of the electrical system and power distribution, and then the implementation of the control software. The final chapters present system testing, evaluation of performance, and concluding remarks."

2 Mechanical Modeling

The complete CAD models are available in the project repository. However, this chapter documents the full modeling process to enable independent development or modification of the design. All components were modeled in Autodesk Fusion 360; nevertheless, any parametric CAD software may be used, as the choice of platform does not influence the resulting mechanical concept.

2.1 Overall Structural Concept

The primary design decision concerns the overall body geometry. Hexapod robots are commonly constructed using either a rectangular chassis with three legs mounted on each side, or a hexagonal body with one leg attached to each face. For this project, a hexagonal configuration was selected. This arrangement provides symmetrical leg distribution, balanced load transfer, and simplified angular spacing between limbs.

2.2 Leg Architecture

Each leg incorporates three actuated joints and therefore requires careful integration of servomotors within the mechanical structure. The design must address three fundamental questions:

- How the leg attaches to the body,
- How individual leg segments are connected,
- How the servomotors are housed while maintaining serviceability.
- How the body segments are connected to allow assembly and access to internal electronics.

A key requirement of the design is maintainability. The structure must allow disassembly for repair or component replacement. For this reason, adhesive bonding was avoided. Instead, each leg segment was divided into two largely symmetrical halves incorporating dedicated fastening features. These features accommodate screws that secure the halves together, ensuring structural rigidity while preserving the possibility of later disassembly.

2.3 Interconnection of Leg Segments

For mechanical stability, each joint between two segments is supported at two distinct contact points.

The first connection point is formed by the servomotor output shaft and servo horn. The servo is mounted within one segment, while the mating segment contains a precisely dimensioned opening through which the output shaft protrudes. The servo horn is then attached externally, clamping the adjacent segment against the motor and transmitting torque directly.

Since standard hobby servomotors provide rotational support only on the output shaft side, a secondary support structure is required to prevent excessive mechanical stress and ensure smooth motion. This is achieved through a protruding cylindrical feature on one segment and a corresponding bore in the mating segment. The circular geometry enables rotational movement while providing lateral stability. In later stages of the project, a sliding bearing is introduced at this interface to further improve motion smoothness and reduce friction.

2.4 Attachment of the Leg to the Body

The connection between each leg and the main body follows the same dual-support principle. The proximal servomotor is mounted directly within a dedicated holder integrated into the robot body. The leg therefore contains two internal servomotors, while the third actuator is structurally supported by the chassis.

As with the inter-segment joints, torque transmission occurs via the servo shaft and servo horn, while an additional protrusion-and-bore interface provides secondary structural support. This interface is also designed to accommodate a bearing element introduced during assembly.

2.5 Body Design

The chassis is modeled as a regular hexagon, with integrated servomotor holders positioned on each face. The internal volume of the body must provide sufficient space for the electrical system installed in later stages of the project. An opening for the main power switch is also included in the body geometry.

The resulting mechanical model satisfies the requirements of symmetry, structural rigidity, serviceability, and integration readiness for subsequent electrical and software implementation stages.

2.6 Body Segmentation

The robot body is divided into two main sections: the lower chassis and the upper cover. These segments are connected using a similar male-female interface as employed in the leg joints, allowing secure attachment while maintaining the ability to disassemble the body. This design facilitates access to the internal volume for installation, maintenance, and adjustment of the electrical components, without compromising structural integrity during operation.

3 Additive Manufacturing

After completing the mechanical design, all components must be exported from the CAD environment in STL format and processed in a slicing software to generate the corresponding G-code for fabrication. The resulting files are then transferred to the 3D printer.

For this project, components were manufactured using the Prusa MK4S and Prusa MINI+

printers. While these machines were selected due to availability and reliability, the design does not depend on a specific printer model. Any fused deposition modeling (FDM) printer with sufficient dimensional accuracy and build volume is suitable.

PETG filament was selected as the primary material. Compared to PLA, PETG offers improved impact resistance and higher temperature tolerance, which are desirable properties for structural components in a robotic system subjected to mechanical stress. The added toughness reduces the likelihood of cracking at screw interfaces and joint regions.

Most standard slicing parameters do not critically influence functionality if reasonable values are chosen. However, infill density has a direct impact on both structural strength and overall mass. Higher infill percentages increase rigidity and durability but also contribute significantly to weight, which in turn affects servo load and power consumption. For operation on flat indoor surfaces, moderate infill is sufficient and preferable to minimize mass. If the robot is intended for more demanding environments, higher infill values are recommended to improve mechanical robustness.

Care should be taken to ensure correct print orientation of load-bearing parts. Components subjected to bending stress should be oriented such that layer lines do not align with the primary stress direction, thereby reducing the risk of delamination. Additionally, dimensional tolerances for servo mounts and bearing interfaces should be verified after printing, as minor deviations may require sanding or adjustment to achieve proper fit.

Proper calibration of the printer prior to manufacturing is assumed, as dimensional inaccuracies directly affect assembly precision and joint alignment.

4 Mechanical Assembly

Following fabrication of all structural components, the mechanical assembly of the robot can commence. The actuation system employed in this project consists of Tower Pro MG996R servomotors, which are discussed in greater detail in the subsequent chapter on electronics. At this stage, the servomotors are treated purely as mechanical elements integrated into the structure.

Each servo is first mounted into its designated housing within the printed components. Proper alignment during insertion is essential to prevent mechanical stress and to ensure that the output shaft remains perpendicular to the mating surface. Servos should be secured firmly using appropriate screws, avoiding over-tightening that could damage the printed mounting features.

Once all servomotors are positioned, the individual leg segments are assembled. The two-part construction of each segment allows the motor to be enclosed and secured using screws, forming a rigid but serviceable joint. Care must be taken to verify smooth rotational movement at each joint before proceeding. Any excessive friction or misalignment should be corrected prior to full tightening of the fastening elements.

After completing the assembly of all six legs, they are attached to the main body structure. The proximal servomotors are mounted directly into the integrated holders within the chassis, forming the final mechanical interface between the legs and the body. At this stage, the structural framework of the robot is complete.

The resulting assembly constitutes the mechanical skeleton of the system, prepared for integration of the electrical components and subsequent system activation.

5 Electrical System Design

6 Software Implementation

The software for the hexapod is written in C++ and designed to run on an Arduino-compatible microcontroller. Development was performed primarily using Visual Studio Code for its advanced code-completion and project management features. Although the PlatformIO library used for uploading code occasionally required multiple attempts to successfully transfer the firmware, Arduino IDE provided consistent uploads at the cost of reduced development features. The choice of IDE does not affect the final functionality of the system.

6.1 Project Organization

The program is modular and separated into five files:

- **Config:** Contains all constants and configuration parameters for the system.
- **Kinematics:** Implements functions controlling the motion of individual legs.
- **Gait:** Uses the kinematics functions to execute coordinated locomotion, specifically a tripod gait.
- **Utils:** Provides general utility functions used across the project.

- **Main:** Handles user input via Bluetooth and invokes the appropriate movement functions.

6.2 Inverse Kinematics

The first computational challenge is inverse kinematics, which translates a desired end-effector position in 3D space into the required joint angles for each leg. For this project, a trigonometry-based solution was implemented due to its simplicity and sufficiency for the workspace and constraints of the robot. This algorithm enables precise leg positioning, allowing higher-level functions to control the robot's movement by specifying target coordinates.

6.3 Locomotion and Gait Implementation

The hexapod employs a standard tripod gait. Legs are divided into two alternating groups of three: two lateral legs on one side paired with the central leg on the opposite side. During motion, one group of legs is lifted and moves forward while the other group remains in contact with the ground, generating thrust and maintaining stability.

Leg trajectories are calculated using vectors representing direction and displacement. User input from a joystick is interpreted as a movement vector, which is combined with the robot's current orientation to determine the target position for each leg. The inverse kinematics functions then compute the necessary joint rotations to execute the desired movement.

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